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Paleomagnetism of Ordovician Lamprophyres from Taylor Valley, Victoria Land, Antarctica

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Abstract – Paleomagnetic study of specimens from four lamprophyric dykes on the Kukri Hills, Taylor Valley (77.64°S, 163.35°E) has yielded a primary mean direction of magnetization of $D = 222.6$ and $I = +0.6$ with $\alpha_{95} = 10.9°$ after AF cleaning. The magnetization of five other dykes and of the amphibolitic basement was either unstable or not fully reliable.

The corresponding pole position lies at 9.3°S and 26.7°E and confirms the previous results from Lower Ordovician rocks from distant areas of East Antarctica.

A Lower Ordovician mean pole position recalculated from valid data lies at 17°S, 21°E.

Key words: Paleomagnetism of Taylor Valley, Antarctica; Antarctica, Paleomagnetism; Ordovician lamprophyres from Antarctica, Paleomagnetism.

1. Introduction

The Dry Valleys located to the west of the McMurdo Sound in the Transantarctic Mts. are the southernmost largest single ice-free area in Antarctica, and therefore have provided unique information about the geology of the continent. A metamorphic-crystalline basement is made by early Paleozoic and older sediments metamorphosed to the amphibolites facies during an Ordovician orogeny (Ross orogeny), accompanied by several subsequent intrusive events of Ordovician age. After a phase of erosion, Devonian to Jurassic sediments were deposited on the peneplanized basement, in mainly continental sequences characteristic of the Gondwana landmasses (Beacon Group). Both basement and continental deposits were subjected to the intrusion of large bodies of Jurassic dolerites (Ferrar dolerites) about 160 m.y. ago.

During late Tertiary times a new cycle of volcanic activity took place with the effusion of the McMurdo volcanics across a large area of the Ross Sea and of the Transantarctic Mts., while a young tectonic cycle affected this area.

In the Dry Valleys and in the neighbouring areas paleomagnetic studies were mostly carried out on the Ferrar dolerites (TURNBULL, 1959; BULL and IRVING, 1960;

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BULL *et al.*, 1962; BRIDEN and OLIVER, 1963) and provided good reference data for the Jurassic position of East Antarctica. Moreover, the studies on basement dykes from the Dry Valleys (BULL *et al.*, 1962) showed that the rock, collected at vertical distances of less than 150 m from the contacts with the dolerite sills, had been remagnetized by the intrusion of the dolerites and directions of magnetization were re-aligned along the direction of the Jurassic field. In fact HASKELL *et al.*, 1965, reported that heating of the host rocks by the dolerite sills could reach 700°C near the contact and 450°C at 70 m away from the contact.

Since new geologic studies (e.g. WARREN, 1969) brought evidence for severe recent faulting in the area with N–S trends and vertical displacements upthrowing the eastern blocks, it could be inferred that a deeper part of the basement outcrops in the eastern area of the Dry Valleys.

We selected this area aiming at the reconstruction of Ordovician directions of magnetization from the McMurdo Sound area.

The field work was carried out in the 1973–74 austral summer during the 2nd Antarctic mission financed by the National Research Council of Italy (CNR) with the free support of the Antarctic Division of D.S.I.R. of New Zealand.

2. *Geology and sampling*

The Taylor Valley is fairly well known from the geological point of view and the abundant literature is quoted and reviewed in a few comprehensive papers on the area: the American Geographical Society geologic map by WARREN, 1969, a paper on reconnaissance geology in the Lower Taylor Valley by ANGINO *et al.*, 1962, a paper with a 1:100 000 map by HASKELL *et al.*, 1965 and finally the work on the stratigraphical relations within the basement by SMITHSON *et al.*, 1970. The basement of the southern flank of the Lower Taylor Valley is mainly made of Cambrian and Precambrian schists of the Skelton Metasediments (schists, quartzites and marbles of Fig. 1), and the local facies recognized by HASKELL *et al.*, 1965 was described as 'the Howard Glacier Belt, 500 ft of metasediments isoclinally folded into two synclines overlying fine-grained granitic rock. The metasediments are interbedded quartz-andesine-amphibole schists, hornblend-andesine-magnetite schists, diopside-vesuvianite hornfels and diopside hornfels'.

The Skelton Metasediments are intruded by a number of plutons and dykes of varying ages, some of which have been dated radiometrically in adjacent areas (MCDOUGALL and GHENT, 1970; PEARN *et al.*, 1963; JONES and FAURE, 1967; GOLDICH *et al.*, 1958; WEBB and WARREN, 1965). The age of the syntectonic intrusives (Olympius granite-gneiss of Fig. 1) and therefore the age of metamorphism is Lowermost Ordovician, estimated at 499 $\pm$ 43 m.y. by FAURE and JONES, 1974, who reviewed critically the previous absolute datings from the region and the stratigraphic sequence of the basement.

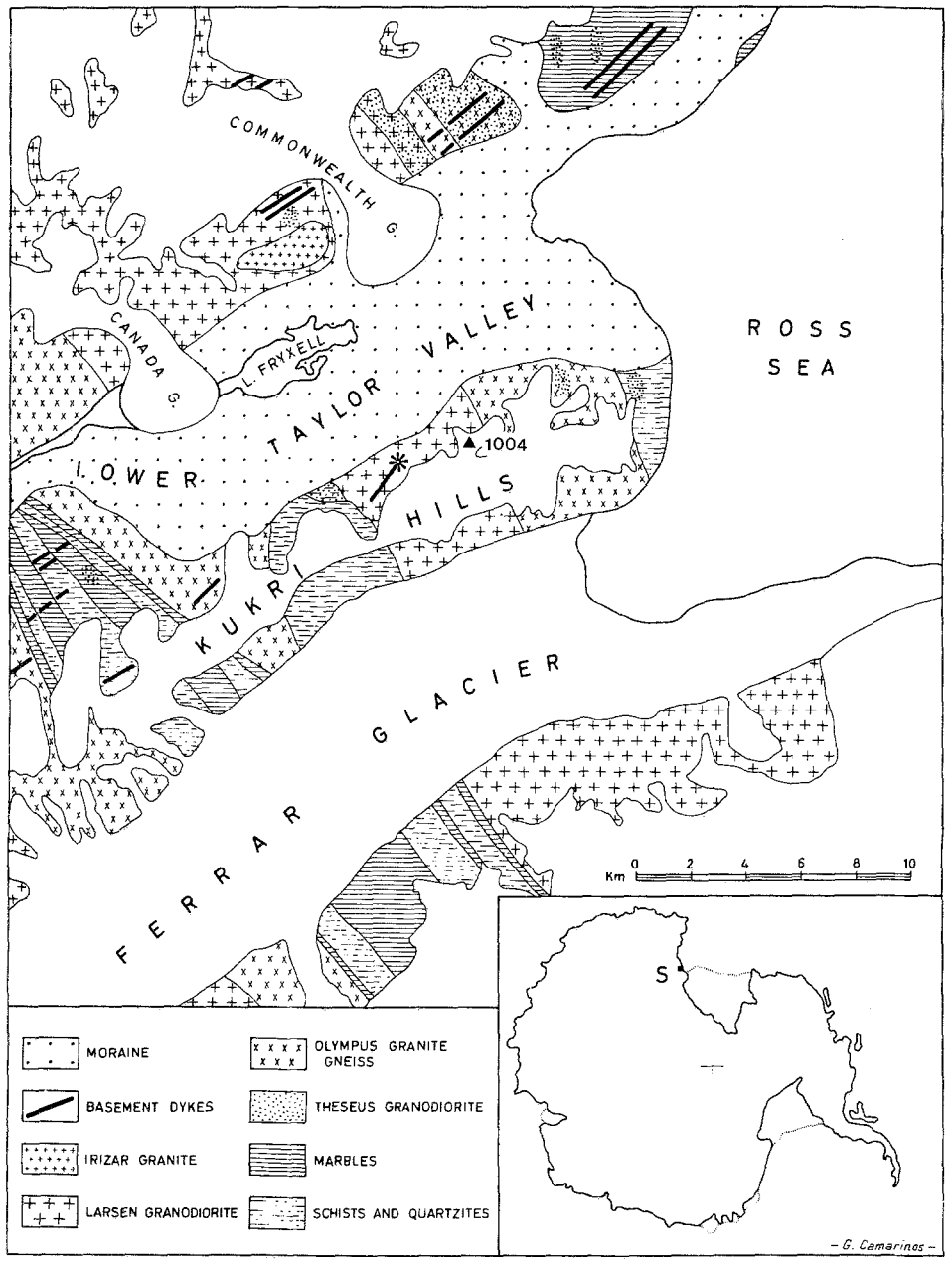


Figure 1
Geologic sketch redrawn after HASKELL *et al.*, 1965, and sampling site (asterisk) of the Ordovician lamprophyric dykes.

The post-tectonic intrusives are subdivided into different units, each with associated sets of acid and mafic dykes, the youngest of which being the Granite Harbour Intrusives (granites and granodiorites of Fig. 1). The basic dykes sampled for paleomagnetic study (basement dykes of Fig. 1) are hornblende lamprophyres described by HASKELL *et al.*, 1965, as the youngest rock within the Granite Harbour Intrusives and therefore locally the youngest rocks of the basement complex. The age is close to 470 m.y., reported for similar dykes from a nearby area by JONES and FAURE, 1967.

The dykes crop out to the south of an E–W striking fault which, according to ANGINO *et al.*, 1962, upthrows the northern flank of Taylor Valley, so that it can be assumed that on the Kukri Hills it is exposed a deeper part of the basement, less likely to have been affected by reheating during the emplacement of the Jurassic sills. In fact, no dolerites are found within 15 km from the sampling area.

The rock is fine grained, with zoned euhedral plagioclase and amphibole phenocrysts in a holocrystalline groundmass. It is held here that this rock is the product of crystallization of microdioritic or perhaps even more mafic magma contaminated by country rock assimilation. Thus in spite of a typical spessartite to kersantite (lamprophyric) mineral composition these rocks should be regarded as microdiorites or amphibolic microgabbros.

The samples were taken from a set of over 30 basic dykes of thicknesses varying from 15 cm to over 2.5 m, striking NE. The similarities of attitude, appearance and petrology between dykes are so close that a synchronous injection can be inferred.

The sequence is located at 163.35°E, 77.64°S between the elevations of 750 m and 900 m, on a lateral ridge of the Kukri Hills about 2.5 km west of the 1004 m hilltop (U.S. Geol. Survey, Antarctica, sheet Taylor Glacier, 1962, 1:250 000 map) (Fig. 1).

Four independently oriented hand samples were taken from each of nine selected dykes, indicated as D1, D2, D3, ..., D9 from south to north. Two additional samples of amphibolitic rock, B1 and B2, were taken from the basement near D9.

The local magnetic declination was measured weekly during the 25 days of stay of the Italian team. During this time the declination remained constant and its correction of 34° west is introduced for all data given in this paper.

3. NRM measurements and magnetic cleaning tests

All tests and measurements have been carried out at the paleomagnetic division of the Laboratorio per la Geologia Marina, C.N.R., Bologna. NRM has been measured on 126 specimens.

Because viscous components of magnetization affecting most readings, the collection was measured again after one month in order to obtain a rough qualitative estimate of the stability of magnetization. According to this test specimens from dykes D1, D3, D4, D7, D9, yielded more stable NRM directions; the samples from the other dykes showed varying degrees of consistency between measurements. NRM

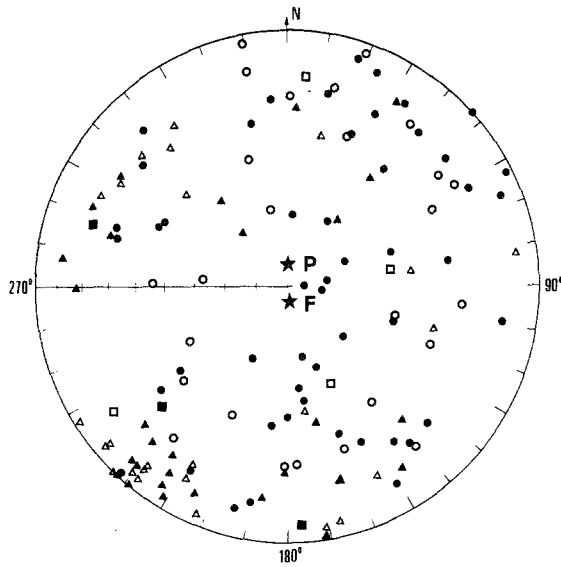


Figure 2
NRM directions of 126 specimens from the amphibolic basement (squares); from the dykes D1, D2, D3, D9 (triangles), and from the dykes D4, D5, D6, D7, D8 (circles). Full (open) symbols: positive (negative) inclinations. P: present geocentric axial dipole; F: present field direction.

directions are widely scattered (Fig. 2). At a closer examination a cluster of directions can be recognized at declinations of about 210–220° with low inclinations; the tests described below will confirm that this cluster corresponds to the stable directions of remanence. Mean directions of each dyke with intensities and precision parameters are reported in Table 1.

Table 1
NRM mean directions from 9 dykes and one site from the basement with precision parameters and intensities

Dykes (site)	N	D	I	α_{95}	R	M (emu/cc)
D1	12	258	+36	54	6.42	1.2×10^{-3}
D2	14	254	+29	57	5.10	1.3×10^{-1}
D3	12	207	+ 1	21	9.93	1.0×10^{-2}
D4	12	57	−25	52	5.42	5.6×10^{-3}
D5	14	262	−65	81	3.30	3.5×10^{-4}
D6	12	39	+25	42	6.59	8.7×10^{-6}
D7	16	205	+45	61	4.84	1.6×10^{-5}
D8	12	156	+81	53	5.30	3.7×10^{-6}
D9	13	226	− 3	20	9.71	6.7×10^{-5}
B	9	—	—	>90	1.14	9.1×10^{-6}

N: number of specimens
D, I: declination, inclination
M: intensity of magnetization
α_{95}, *R:* precision parameters

Intensities of NRM of the dykes D1, D2, D3, D4 are of the order of 10^{-1} to 10^{-3} emu/cm^3, while the NRM intensities are very low for dykes D6, D8, D9 and for the basement (10^{-5} to 10^{-6} emu/cm^3) and intermediate for D5 and D7.

Seventy-six specimens have been demagnetized stepwise, in AF up to 2750 Gauss max. field peak value along three axes, and measured. The patterns of directional changes and intensity variations are quite different from dyke to dyke and are more consistent for the dykes with higher intensities of NRM, corresponding also to higher magnetite contents observed in thin sections.

A set of diagrams of decay of intensities with the corresponding directional changes is shown in Fig. 3.

A remarkable feature of several decay curves is that the decrease of intensities is not as regular as it generally is for rocks suitable for paleomagnetic study: Fig. 3d is an example of an irregular curve from dyke D5. Any explanation for such a behaviour does not bring favourable evidence for stability or even for existence of remanence, and in fact the data from those dykes had to be discarded.

However more regular patterns of decay curves were obtained from some dykes: the magnetization of dykes D1 and D2 decreases with some regularity with increasing AF peak fields. The large directional changes show that strong soft secondary components of magnetization are present. The directions of magnetization of pilot specimens from dyke D1 reach an area of relative stability after 100 Gauss peak field and become scattered again at all steps over 800 Gauss (Fig. 3a).

Similar situations of marked changes of directions of magnetization with cleaning up to 200 Gauss are plotted for pilot specimens of the dyke D2 in Fig. 3b. The scatter of directions over 800 or 1000 Gauss is due to viscous components after demagnetization by high alternating fields.

Most specimens from the dyke D9 follow a similar pattern of changes of directions of magnetization, with the sole difference that an area of relative stability and the best cluster of directions of magnetization are attained after cleaning at lower AF fields, between 50 to 100 Gauss.

The behaviour of the directions and intensities of magnetization for specimens from the dyke D3 is shown in Fig. 3c. In this dyke secondary components of magnetization are not present or do not appear to affect significantly the directions of remanence; as in the other cases, the stable component cannot be measured after demagnetization at 800 to 1000 Gauss.

The cleaning tests on specimens from the basement show characteristics of magnetization similar to those of the dykes D1 and D2 with the only difference that magnetization becomes unstable after applied fields of 400 Gauss. Diagrams are found in Fig. 3e.

The demagnetization tests of specimens from the dykes D4–D8 show that a stable and reliable remanent magnetization is not present because of several unfavourable conditions. Directions as well as intensities change widely and irregularly with increasing demagnetization. Clusters of specimen directions with α_{95} lower than

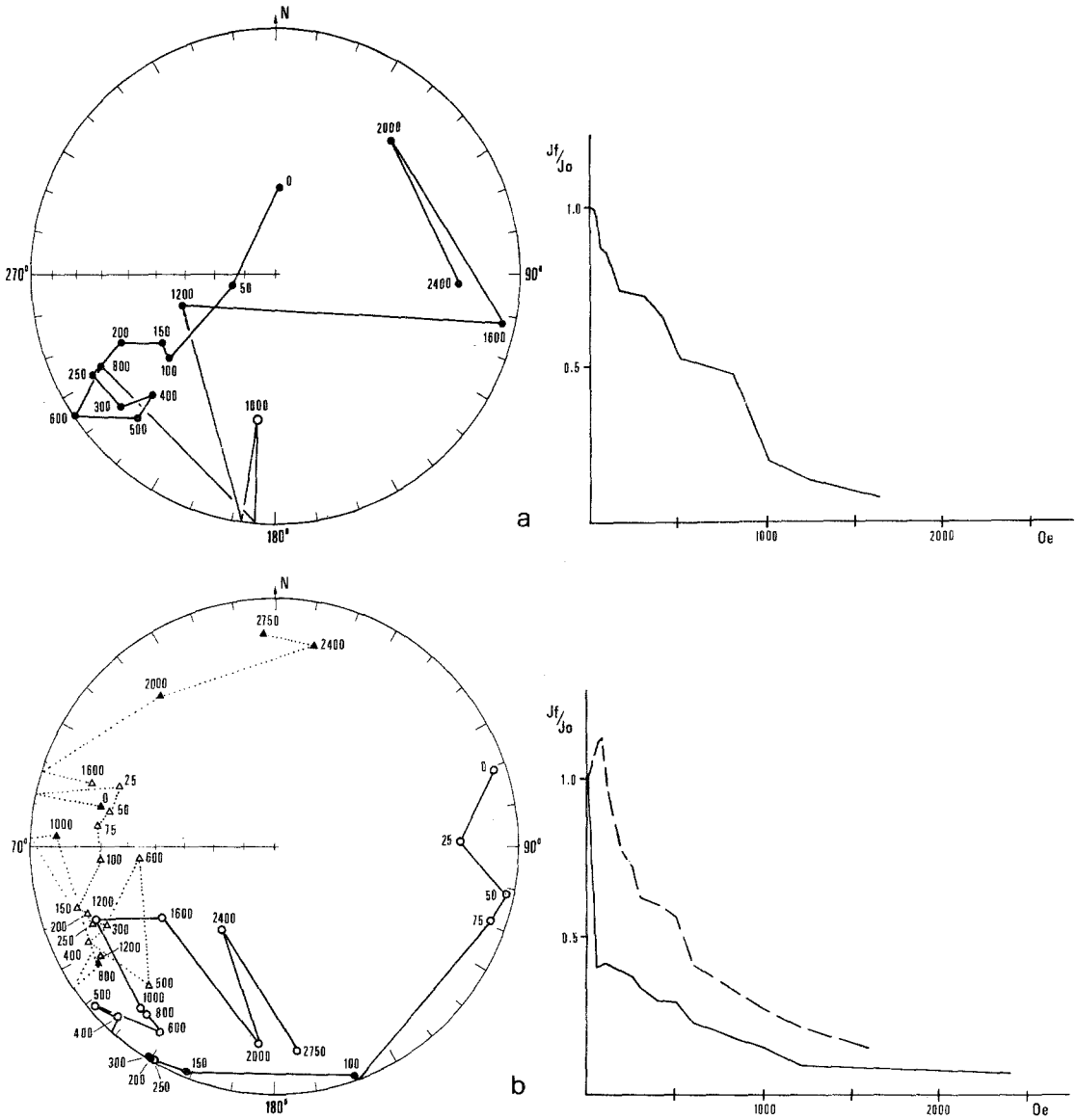


Figure 3

Directions and decay curves of intensity of magnetization with progressive AF cleaning. Stereograms: directions of magnetization after stepwise cleaning (0 to 2750 Oersted). Diagrams: demagnetizing fields in abscissae (Oersted).

3a: 1 specimen from dyke D1
3b: 2 specimens from dyke D2
3c: 2 specimens from dyke D3
3d: 2 specimens from dyke D5
3e: 1 specimen from the basement

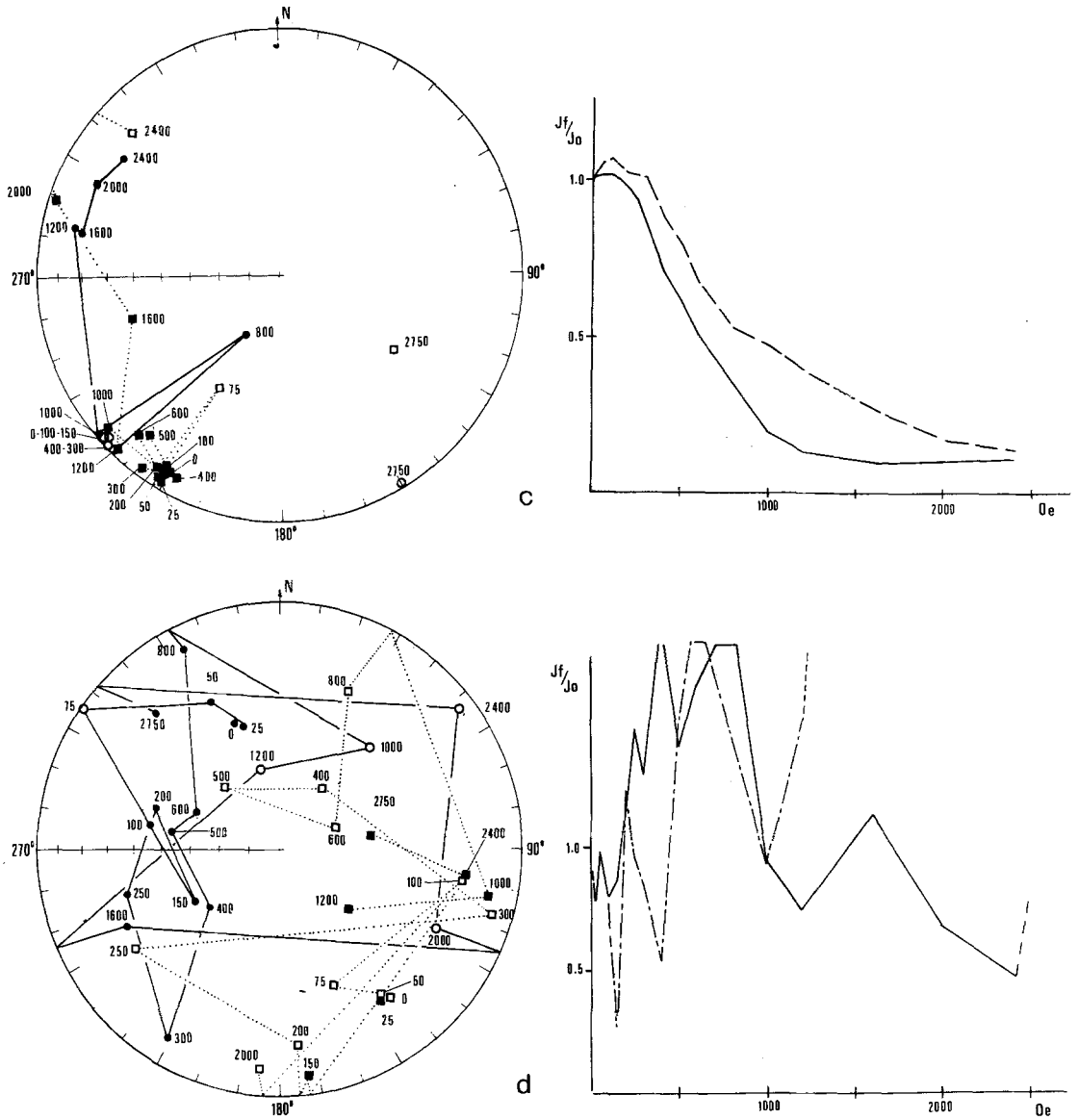


Figure 3 – *continued*

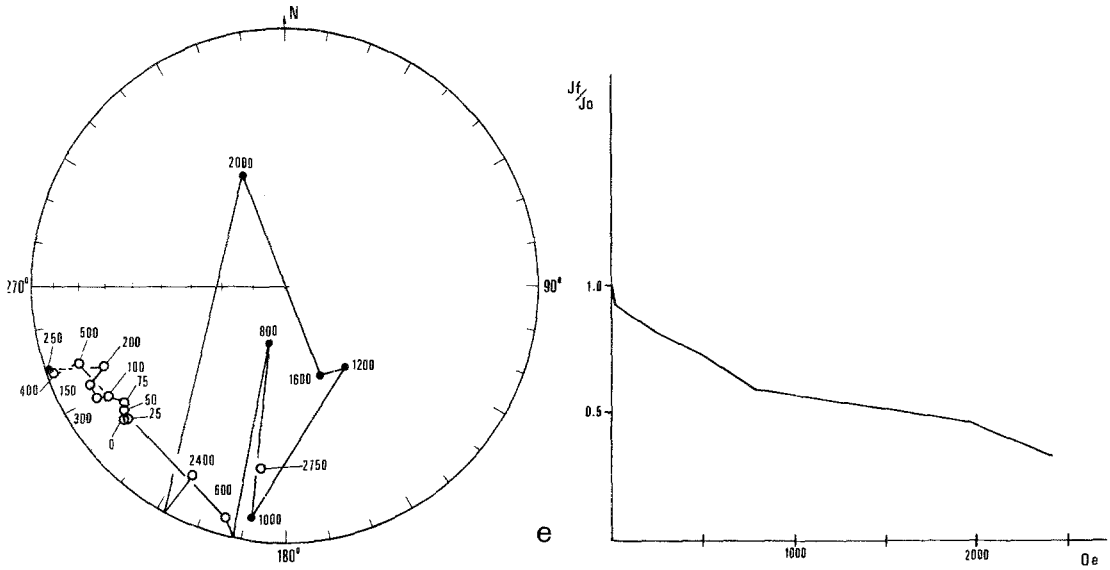


Figure 3 – *continued*

40° (from 8 specimen directions for each dyke) were never attained at any step of demagnetization. Semiangles of circles of confidence α_{95} often exceed 90° and are larger than the correspondent α_{95} semiangles of NRM measurements. Frequently specimens cut from the same core yield totally different directions. Intensities do not decrease regularly with progressive cleaning. Strong viscous components of magnetization varying rapidly with time made the measurements unreliable, even when a recording instrument was used.

For these reasons we have not considered it necessary to report data and diagrams, but for the example of Fig. 3d, and the samples from dykes D4–D8 were discarded as unreliable for further paleomagnetic studies.

It should only be remarked that the dykes discarded are those with lowest content in magnetite (see Petrographic note, p. 975).

4. Results

On the basis of the demagnetization tests and of the groupings of directions of remanence found for some of the lamprophyric dykes, we consider the data from D1, D2, D3, D9 and from the basement B to be reliable.

Table 2 reports data from these rocks for the cleaning fields which have proved to reveal directions of magnetization within the range of relative stability, i.e. after cleaning between 100 and 500 Gauss, in order to allow direct comparison.

Results are also shown in stereographic projection in Fig. 4, where all specimen

Table 2

NRM and after cleaning mean directions from 4 dykes and basement, and overall mean direction of dykes with stable remanent magnetization

Dykes (Site)	N	NRM			100			200			300			400			500		
		D	I	α_{95}	D	I	α_{95}	D	I	α_{95}	D	I	α_{95}	D	I	α_{95}	D	I	α_{95}
D1	12	258	+36	54	240	+86	44	232	+13	34	230	+8	13.5	231	+12	12	229	+8	12
		$R = 6.42$															$R = 11.22$		
																	$M = 5.8 \times 10^{-4}$		
D2	14	254	+27	57	244	+10	23	231	+2	13	228	−3	10	226	−2	10	228	−2	10
		$R = 5.10$									$R = 13.29$								
											$M = 1.0 \times 10^{-3}$								
D3	12	206	+1	21	211	+2	11	219	+2	7	218	+1	7	219	+3	6	214	+4	7
		$R = 9.93$												$R = 11.76$					
														$M = 1.2 \times 10^{-4}$					
D9	13	225	−3	20	220	−1	23	222	−20	37	214	−9	14	214	−5	15	214	+3	20
		$R = 9.71$				(6)			(6)					$R = 11.57$					
														$M = 6.5 \times 10^{-5}$					
B	9			90	203	+ 2	54	248	+2	74	252	−1	33	253	−4	23	218	+12	75

Overall mean: $N = 4$, $D = 222.6°$, $I = +0.6°$, $R = 3.96$, $\alpha_{95} = 11$.

N: number of specimens (sites for the overall mean).
D, I: declination, inclination.
α_{95}, *R:* precision parameters vive!
M: intensity of magnetization (emu/cc units).
Figures within brackets: number of specimens when different from *N*.
Overall mean is calculated from cleaning fields of 500 Oe for D1, 300 Oe for D2, 400 Oe for D3 and D9.

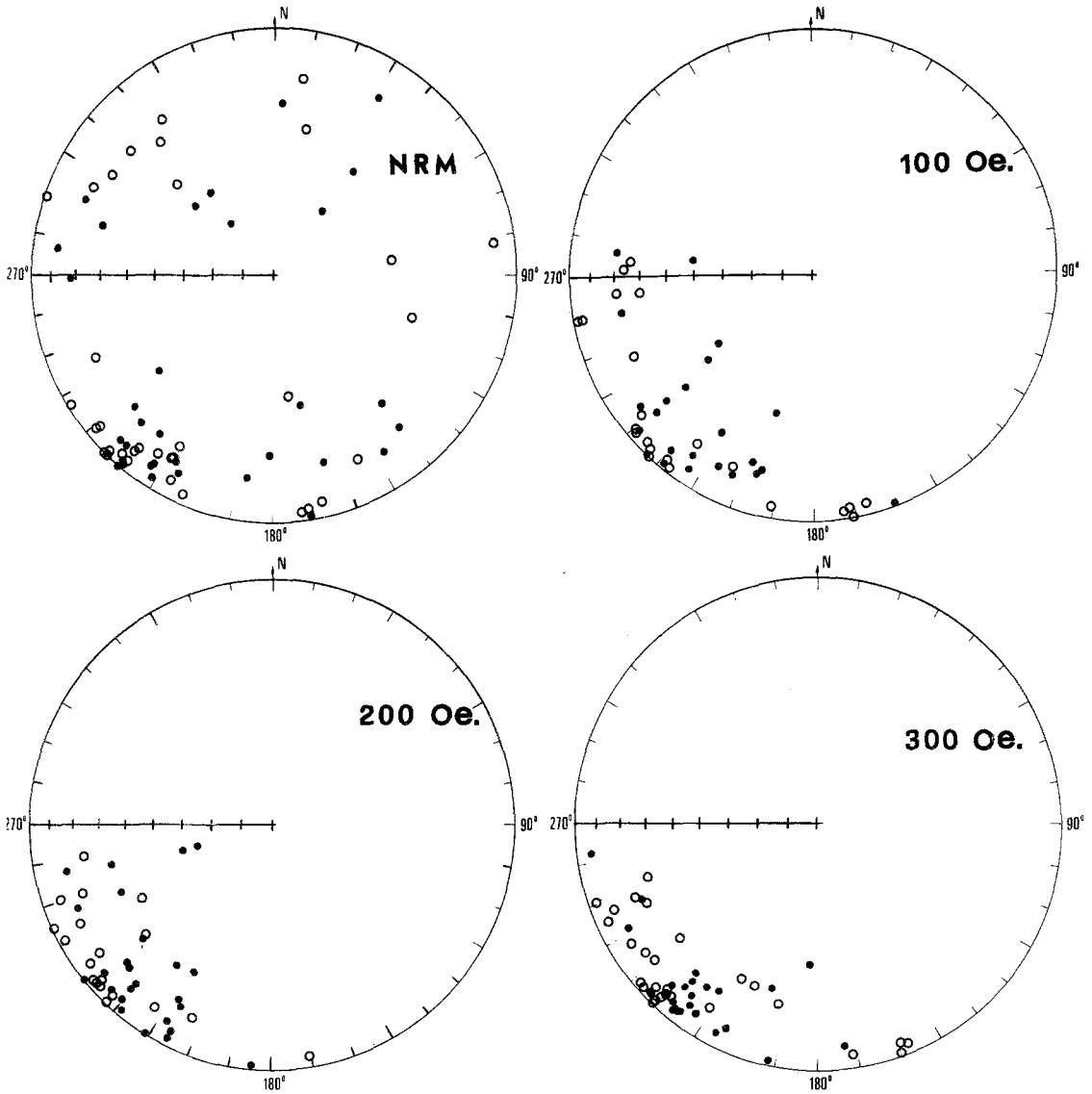


Figure 4
Directions of magnetization of all specimens from dykes D1, D2, D3, D9, NRM and after cleaning at 100, 200, 300, 400 and 500 Gauss max. peak fields.

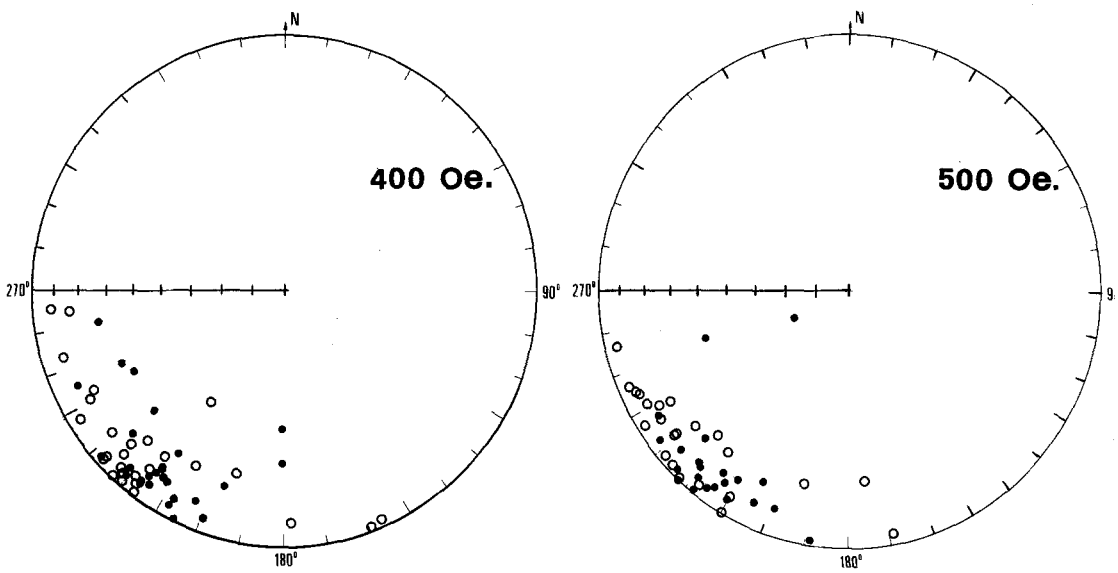


Figure 4—*continued*

directions of the four dykes are plotted, and in Fig. 5, with the accepted mean directions of the four dykes and the overall mean direction; for comparison also the Jurassic, the present dipole and the present field directions are shown. From the sequence of stereograms of Fig. 4 it is evident that directions of magnetization cluster progressively toward southwesterly declinations and low inclinations.

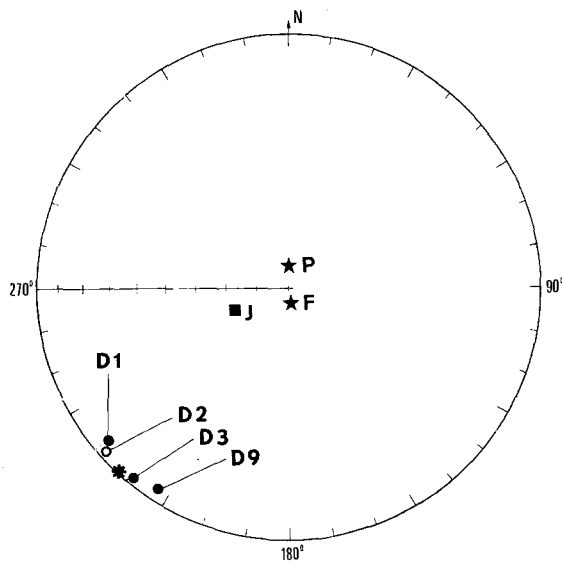


Figure 5

Mean directions of magnetization after AC cleaning at the chosen fields for dykes D1, D2, D3, D9 and overall mean from the four dykes (asterisk). P: present dipole; F: present field; J: Jurassic field direction.

We will remark here that a residual scatter of directions, still observed for some specimens after cleaning at 300 and 400 Gauss, lies along a great circle of low inclinations quite close to the horizontal plane. We have discarded these few scattered data from the computations of the average directions of Table 2, assuming that they may well represent directions of specimens insufficiently cleaned from a secondary component of magnetization. Several specimens have followed a path similar to the one shown in Fig. 3b during the stepwise demagnetization tests.

For the computation of the pole position, mean directions of magnetization after cleaning at 300 Gauss were chosen for the dyke D2 at 400 Gauss for dykes D3 and D9 and at 500 Gauss for the dyke D1 (Fig. 5 and Table 2).

The overall mean direction of $D = 222.6$ and $I = +0.6$ is computed by giving unit weight to each dyke. Such procedure might not be necessary in view of the geologic considerations of the closeness of synchronously injected dykes. In fact the α_{95} parameters are not significantly different whether computations are carried out from specimens' directions or from hand-samples' directions.

The data from the amphibolitic rock of the basement is excluded from the computation of the pole position because it is based on few specimens and because its age must be older than that of the dykes, although still within the Ordovician.

The pole position computed from the directions of the lamprophyric dykes from the Kukri Hills lies at 9.3°S and 26.7°E with polar errors dm 10.9° and dp 5.5°.

5. *Discussion and conclusion*

It is generally agreed that East Antarctica represents the ancient crustal unit that once belonged to Gondwanaland and its pole positions derived from paleomagnetic data are used for the reconstructions of the southern landmass (e.g. McElhinny and Luck, 1970).

Paleomagnetic results from West Antarctica from rocks of varying ages give pole positions displaced from those of East Antarctica and from other southern continents (e.g. Le Roy Scharon *et al.*, 1969, Schopf, 1969), since West Antarctica was subject to a different history of mountain building and drift (e.g. Hamilton, 1967). Therefore we will limit the comparison of the data from the lamprophyric dykes of Taylor Valley to other Paleozoic data from East Antarctica only.

The polar path for East Antarctica is not very detailed: apart from preliminary data from 9 samples of Precambrian dolerites about 1000 m.y. old from the Vestfold Hills by Embleton and Arriens, 1973, two valid poles are known for the Paleozoic: a Cambro-Ordovician pole at 1.5°S and 28.5°E from charnockites about 502 m.y. old from Mirnyy Station (McQueen *et al.* 1972), and a Lower Ordovician pole at 28°S and 10°E from intrusives from the Sør Rondane Mts. dated to 480 m.y. (Zijderveld, 1968).

Pole positions of later ages are known for the Jurassic and for Cenozoic times.

Although scarce, these pole positions allow the fit of the main Antarctic landmass at its place in the reconstructions of Gondwana: e.g. McELHINNY, 1973.

Data from Devonian rocks by NAGATA, 1969, are not often considered to be valid because the samples have not been demagnetized, and so are other previous data from Paleozoic formations from Victoria Land, since the authors themselves observed that the rocks had been remagnetized in Jurassic times (TURNBULL, 1959; BULL *et al.*, 1962).

Our pole position at 9.3°S and 26.7°E (this southern pole position is shown in Fig. 6 and compared with other data), derived from rocks of about 470 m.y., lies in

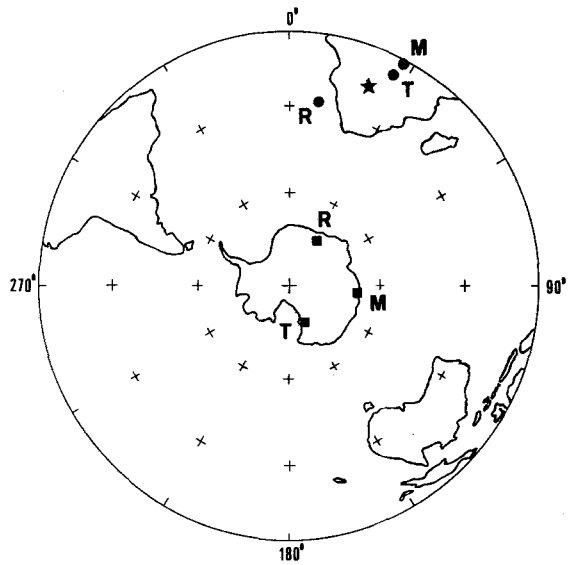


Figure 6
Pole positions (dots) for Antarctica from Ordovician rocks from Mirnyy Station (M), Sør Rondane Mts. (R) and Taylor Valley (this paper, T); new average Ordovician pole position (star) and sampling sites (squares).

between the pole positions from Mirnyy station and Sør Rondane Mts., over 2000 and 3000 km away from this sampling site respectively, and can well be considered to confirm the previous data.

By giving unit weight to each of the three pole positions the average Lower Ordovician pole for East Antarctica is calculated at 17°S and 21°E.

A speculative remark could be proposed on the relative latitudes of the three poles: the latitude of the pole position calculated for the older Cambro-Ordovician charnockites from Mirnyy Station is lower than the two other Ordovician pole latitudes. Although this difference might be significant and might suggest a phase of drift toward the south of about 15° during the Lower Ordovician, such a hypothesis

needs to be controlled by further evidence of two orders: firstly, since East Antarctica supposedly belonged to Gondwana during the Ordovician, a consistent synchronous polar drift should be recognized from other southern continents; secondly more Cambrian and Ordovician data should be gathered from a single area of reference in East Antarctica. Otherwise no component of drift or of polar wandering could be easily distinguished from the likely geotectonic motions occurring within Antarctica during the Ordovician orogeny.

Acknowledgements

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Petrographic note

This petrographic description is due to A. ALBERTI (Institute of Mineralogy and Petrology of the University of Trieste, Italy).

The dyke samples are homogeneous and have isotropic texture, except for a few granite and schists xenoliths occurring in some cases. They consist of subequal amounts of mafic and light minerals, identifiable as tiny feldspar laths (1 to 2 mm long). The texture is microporphyritic seriate with zoned euhedral plagioclase and more or less biotitized, elongate amphibole phenocrysts.

In some rocks the groundmass consists of a micrographic intergrowth of acicular amphibole and zoned feldspar. In all rocks, plagioclase and amphibole occur in subequal amounts. Amphiboles in glomeroporphyritic associations may occasionally show a relic core consisting of clinopyroxene, probably augite.

A most interesting feature of the dykes in thin sections is the occurrence of irregularly shaped amphibole aggregates or clots of variable size (but never exceeding a few mm in diameter), which may coexist with the amphibole microphenocrysts. They are irregularly scattered throughout the rock, conferring to it a marked heterogeneity (which is not, however, noticed in the hand specimens). These clots have already been observed by GHENT and HENDERSON, 1968, but do not clearly appear to have the regular succession of zones (as for form, grain size and composition) mentioned by the same authors.

Zoned anhedral plagioclase, amphibole, minor alkali feldspar and chlorite are the main components of the ground-mass.

As for mineral composition, plagioclases range from bytownite-labradorite (70–80 An) to oligoclase (20–30 An), generally with strong normal zoning; the calcic core mostly shows discontinuous sericitic (rarely calcitic) alteration. Biotitization appears to be accompanied by sphene development, but small anhedral sphene specks also occur sparsely in the rock fabric. The larger biotite pseudomorphs may show sagenitic aggregates.

Accessories are aptite, sphene and magnetite as the sole opaque phase. Allanite has been recognized in one sample. Magnetite is weakly oxidized showing a very thin hematite rim; it often occurs in small octahedra. Variable amounts are observed in the dykes: very abundant in D1, in small scattered octahedra, and also in D2 and D3, with few large allotriomorphic grains; it is scarce in D5, D6 and D7 with small grains within the amphibolite clots, and almost absent in D4 and D8. The magnetite grains are found within or close to the amphibolite clots thus suggesting an origin by iron release during the biotite and amphibole pseudomorphism after pyroxene; they often are surrounded by small magnetite flakes.

The dyke D9 shows the association plagioclase-brown horneblende with no biotite pseudomorphs. It is therefore a spessartite and can be considered the only true lamprophyre. The mafic minerals show evidence of progressive hydration, going from clinopyroxene to amphibole and to biotite. The amphibole aggregates have been explained by GHENT and HENDERSON, 1968, either as disequilibrium crystallization of microdioritic magma or as reaction between the same magma and tremolite xenoliths. Owing to the occurrence of small, partly resorbed quartz xenocrysts (schist and granite xenoliths have also been noticed in the field), the idea of a more mafic magma, modified by granite, perhaps even by marble or lime and quartz-rich schist assimilation, can be entertained.

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